Predicting Acute Food Insecurity with Machine Learning

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Abstract

Food crises arise from interacting drivers—conflict, poverty, climate variability and economic shocks—producing acute food insecurity (AFI) among vulnerable populations. Early warning systems (EWS) depend heavily on the Integrated Food Security Phase Classification (IPC), the gold standard for assessing AFI severity and widely used across the humanitarian system. Yet IPC assessments are resource-intensive, infrequent and especially difficult in conflict-affected settings, constraining timely decisions. We develop a machine-learning model that leverages passively collected data to predict IPC phases and estimate populations facing high AFI (IPC Phase 3+). Trained on multi-source indicators (e.g., prices, infrastructure, conflict intensity, vegetation, night lights and weather), the model correctly identifies 94.1% of food crises and attains a 77.5% success rate for crises predicted 12 months ahead. We generate contemporaneous estimates, nowcasts and forecasts, providing a scalable complement to IPC with the potential to strengthen EWS and support more timely, data-driven humanitarian action.

Keywords: Food security, Food crisis, IPC, Predict IPC

Main

Food crises result from the complex interplay of conflict, poverty, climate variability, extreme weather events, and economic shocks. They are marked by high and rapidly increasing levels of acute food insecurity (AFI) among vulnerable populations [1]. Early warning systems (EWS) capable of anticipating food crises in a timely manner are essential to guide humanitarian assistance and mitigate the worst impacts of major disruptions in food access.

The Integrated Food Security Phase Classification (IPC) system is a multi-partner initiative designed to enhance food security and nutrition analysis and serves as a key resource for early warning and early action mechanisms to anticipate food crises [2]. The IPC scale classifies vulnerable populations into five phases based on the severity of AFI in a geographic area: 1) Minimal, 2) Stressed, 3) Crisis, 4) Emergency, and 5) Catastrophe [3]. A food crisis is defined as a situation in which at least 20% of the population is classified in Phase 3 or above and requires urgent humanitarian assistance.

Donors consider the IPC as the gold standard for assessing the severity and scope of AFI within countries and for guiding the development of targeted response plans [2]. However, IPC operations for data collection and analysis are costly, as they rely on both qualitative and quantitative data, including on-the-ground assessments conducted by multiple agencies. These efforts are often hindered by challenging conditions for collecting high-quality data, especially in conflict-affected areas. Due to high costs and limited budgets, IPC assessments are not always timely or frequent and may lack coverage in many countries identified as at risk of food crises [4].

Moreover, food insecurity in fragile contexts can deteriorate rapidly—sometimes within weeks—as illustrated by recent crises in the Gaza Strip and Sudan [5, 6]. These cases highlight the urgent need for timely evidence to inform decision-making, as well as the importance of more frequent, ideally real-time, monitoring of food insecurity. While direct, real-time measurement of food security remains difficult, growing availability of passively collected data enables monitoring of key risk factors. Leveraging such data can enhance the predictive capacity of EWS, facilitating earlier and more effective responses to emerging crises.

Recent studies have demonstrated the potential of machine learning (ML) models using passively collected data to predict poverty, malnutrition, and food insecurity at relatively low cost [7–10]. Building on this literature, we develop an ML model that integrates passively collected indicators to predict IPC phases and estimate the proportion of the population facing food crisis.

Our model forecasts AFI severity up to 12 months in advance, while also supporting nowcasting and contemporaneous estimation. We construct a dataset comprising 5,575 IPC observations across 1,198 subnational areas in 29 countries from 2017 to 2022, enriched with geospatial, socioeconomic, and environmental covariates such as food prices, weather conditions, conflict events, vegetation indices, nightlight intensity, and market access.

When tested against published IPC assessments, our forecasting model correctly identifies 94.1% of food crisis cases, while 77.5% of cases predicted to be in crisis are confirmed as such 12 months later. The nowcasting and contemporaneous models show

similar or stronger predictive performance. Across all models, key predictors include food price indicators, the infrastructure index, conflict-related variables, vegetation indices, nightlight intensity, and weather conditions—underscoring the complex and multidimensional nature of food crisis risk. These results emphasize the value of integrating diverse data sources to strengthen early warning and support more timely, evidence-informed responses.

Our paper makes two key contributions. First, our model estimates both the overall IPC phase and the population share in each phase.[1] While most existing studies focus on related well-being indicators—such as asset ownership, child malnutrition, or food consumption scores [9–12]—only a few, including [7] and [9], attempt to predict IPC-like outcomes. Unlike these studies, which rely on the classifications derived from the Famine Early Warning Systems Network (FEWS NET) and focus on a binary transition into crisis (Phases 3-5) from Phases 1-2, we use the IPC assessment outcomes themselves—the consensus estimates produced by IPC's analytical process and partner agencies. This choice makes our approach both methodologically distinct and directly relevant to early-warning and early-action systems that rely on IPC assessments.

An additional advantage of the IPC data for our purposes is that the assessments provide not only an overall phase classification for each area but also estimates of the population in each phase. This allows us to predict the share of the population experiencing food crisis, an outcome that is highly relevant for assessing the potential scale of humanitarian need. At the same time, IPC assessments are conducted irregularly, and about 43% of areas are observed only once or twice during the 6-year study period (see Supplementary Fig. 1). This uneven longitudinal structure creates additional challenges for predictor construction, model training, and prediction. It also makes transition-based modeling difficult in our setting, because many areas lack sufficient consecutive assessments to construct transition outcomes consistently across locations.

Second, we provide estimates across three temporal horizons: contemporaneous, nowcasting, and forecasting. Contemporaneous estimates can help fill gaps in historical IPC assessments for areas covered by IPC at other points in time. Nowcasting provides updated assessments using the most current available data, while forecasting offers lead time for early action. By evaluating food crisis risks across all three timeframes—past, present, and future—our model can help humanitarian and development actors better prioritize and time their interventions.

Results

In this section, we begin by presenting the results of our main model—the forecasting model—followed by the nowcasting and contemporaneous results. We then compare model performance with simple baselines and explore the identification of more severe Phase 4+ cases, alternative geospatial modeling strategies, and the model's ability to generalize to new geographic areas.

[1] The IPC classifies populations within a given area by severity of AFI using Phases 1 through 5. It also assigns an "overall IPC Phase" to the area based on the average condition of the population, applying a 20% threshold criterion. See the "Data" section for further details.

Forecasting

In this model, the training period spans January 2017 to December 2021, with forecasts generated for January to December 2022. For each IPC assessment conducted in 2022, we use predictors available 12 months earlier to forecast the overall IPC phase and the percentage of the population in each phase one year in advance (see the Methods section for details on IPC phase classification).

Prediction Performance. The model's performance is summarized in Panel A of Table 1 and illustrated in Fig. 1. When the outcome variable is the percentage of the population experiencing food crisis, the model achieves a moderate out-of-sample (OOS) R^2 of 0.249. While this indicates limited ability to explain variation in the continuous outcome, 12-month-ahead prediction of noisy socioeconomic outcomes is inherently challenging.[2] To illustrate how these predictions could nevertheless be useful for prioritization, we use the predicted share of the population in food crisis to identify higher-risk locations. Supplementary Fig. 2 highlights areas in the top 30% of the predicted population share in food crisis in 2022 and compares them with areas in the top 30% based on the observed population share. The two maps show broadly similar geographic patterns. This exercise illustrates how the model could potentially serve as a screening and prioritization tool to identify locations warranting further monitoring and assessment, rather than as a precise estimator of the affected population.

For IPC phase classification, the overall accuracy is 65.0%, indicating that the model correctly forecasts IPC phases in 65.0% of test cases. Focusing specifically on food crisis conditions, the model achieves a sensitivity (recall) of 94.1%, meaning it correctly identifies 94.1% of all food crisis cases. The precision is 77.5%, indicating that 77.5% of predicted food crisis cases actually materialized after one year.

Although food crises often exhibit a protracted nature [14], we aim to ensure that the model's classification performance for food crisis is not simply driven by this temporal stability or by reliance on static predictors. If performance were primarily due to the persistence of food crises, we would expect the model to perform poorly on instances where the IPC phase changes over time. To test this, we excluded from the test set all observations in which the IPC phase matched that of the most recent training round. The model's performance remained strong—with an accuracy of 0.593, sensitivity of 0.990, and precision of 0.702 (Panel A of Table 1)—indicating that it captures meaningful dynamic signals rather than merely reflecting temporal inertia.

Feature Importance. To further analyze our model and better understand the role of individual predictors in forecasting food crises—thereby enhancing its policy relevance—we apply SHAP (Shapley Additive Explanations) values to assess feature importance, following [15]. SHAP values quantify each feature's contribution to the prediction by averaging its marginal effect across all possible feature combinations. Panel A of Fig. 2 displays the feature importance rankings for forecasting IPC phases.

[2] Similarly moderate OOS R^2 values have been reported in related applications. For example, Constenla-Villoslada et al. [13] report an average OOS R^2 of 0.12 for 12-month-ahead forecasts of global acute malnutrition prevalence at the ward level using machine-learning models based only on secondary data and without lagged outcomes, as in our setting.

For predictions of IPC Phase 3 or above, WFP food price spatial dispersion ranks first and month-on-month price change ranks fifth in SHAP importance (Fig. 2, Panel A, top). The infrastructure index is also highly influential, emphasizing the role of local infrastructure in shaping food security outcomes. Environmental and economic proxies such as vegetation indices, soil moisture, and nightlight intensity further contribute meaningfully to model performance, underscoring the value of dynamic features in anticipating food insecurity. The importance of these dynamic predictors reinforces our earlier finding that the model is indeed learning from evolving, rather than static, inputs.

Geographic coordinates rank relatively highly in the feature-importance analysis, which is expected given that certain areas are persistently vulnerable to food crises due to conflict, climate shocks, or other structural factors.[3] This result, consistent with the protracted nature of food crises documented in [14], further supports the need for geographically targeted and context-specific interventions.

When predicting IPC Phase 4 and above levels of food crises, the previously discussed factors remain relevant; however, variables such as temperature, shortwave radiation, and soil pH gain influence. This shift suggests that weather shocks and underlying environmental conditions are strongly associated with severe food crises, although given the deployed methodology we cannot affirm causality.

Fig. 2 Panel C presents the feature importance of grouped variables across IPC phases, with predictors categorized into six domains: physical geography, weather, conflict, agricultural and land productivity, food prices, and other economic conditions. The figure reveals distinct patterns in the relative importance of these groups as food crises intensify. For predictions of Phase 3 or above, food price indicators are particularly influential, with other economic variables also playing an important role.[4] When forecasting Phase 4 or above, food price factors remain significant, but the contributions of conflict and weather-related variables become more pronounced. These findings are consistent with IPC assessments identifying conflict, extreme weather events, and economic shocks as important drivers of acute food insecurity and underscore the value of monitoring areas affected by violence, climatic shocks, and food-market volatility (see, e.g., [14]).

Nowcasting

As with the forecasting model, we train the nowcasting model using IPC data from 2017 to 2021 and reserve 2022 data for testing. However, the nowcasting approach differs by incorporating updated predictors—specifically, those available at the time of the 2022 IPC assessments. This allows the model to leverage contemporaneous information to more accurately estimate current food insecurity outcomes.

[3] Our robustness checks show that excluding latitude and longitude produces little change in predictive performance. This suggests that other spatially structured covariates capture much of the geographic information represented by the coordinates (see the "Alternative Geospatial Modeling Strategies" subsection in the Results section).

[4] Food-price spikes and spatial price dispersion should be interpreted as potential signals of underlying disruptions, which may be caused by factors such as conflict, weather and production shocks, weak infrastructure, poor market connectivity, and broader macroeconomic instability. They should not be taken as evidence that food inflation as such is the main cause. Greater attention should therefore be devoted to identifying the context-specific forces that generate damaging price instability and to assessing policies that address those underlying causes, rather than focusing solely on direct price stabilization.

The nowcasting framework adopts a cascading two-layer structure. The first layer uses all forecasting features from secondary data observed 12 months prior to each IPC assessment in the test set. The second layer enhances these predictions by incorporating additional features from data observed during the 12 months leading up to the assessment, thereby capturing residual variation not explained by the initial layer.

Prediction Performance. The prediction performance of the nowcasting model is summarized in Panel B of Table 1 and illustrated in Fig. 1. The model achieved an accuracy of 0.654, a sensitivity of 0.942, a precision of 0.777, and an out-of-sample R^2 of 0.253. Even after excluding test observations that fall within the same IPC phase as the final round of the training set, the model maintains similar performance (accuracy: 0.571; sensitivity: 0.964; precision: 0.733), consistent with the results from the forecasting model.

A comparison between nowcasting and forecasting models yields a key insight: the performance gains from incorporating contemporaneous predictors in the nowcasting model are modest. This is likely because many of the risk factors that contribute to food crises emerge well in advance of the actual transition into severe phases. As a result, lagged predictors remain highly predictive and effective for forecasting IPC phases 12 months ahead—underscoring their practical value for early warning and proactive response systems.

Feature Importance. As shown in Panel B of Fig. 2, the feature importance results from the second-layer nowcasting model closely mirror those of the forecasting model (top figure of Panel A), with WFP food price spatial dispersion and temporal changes emerging as the most influential predictors. Nightlight intensity also plays a key role in predicting IPC Phase 3 or higher.

Furthermore, SHAP values of predictor categories from the second layer of the nowcasting model, shown in Panel D of Fig. 2, confirm that food price shocks remain the dominant predictor for Phase 4 or higher classifications, while other economic conditions gain increasing importance for predicting Phase 3 and above. Taken together with the feature importance results from the forecasting model, these findings underscore the critical need for real-time monitoring of local market food prices and nightlight intensity to enable timely detection of emerging food crises and inform effective early responses.

Contemporaneous Prediction

In the contemporaneous model, we perform a cross-sectional analysis using random five-fold, row-level cross-validation, following the method of [11]. The 5,575 observations are assigned to five equal folds with a fixed random seed of 0. Each iteration uses 4,460 observations for training and 1,115 for testing; performance metrics are computed from the pooled out-of-fold predictions covering all 5,575 observations. The contemporaneous model achieves an accuracy of 0.693, sensitivity of 0.904, precision of 0.797, and R^2 of 0.642.

As expected, the contemporaneous model exhibits a substantially higher R^2 than the nowcasting and forecasting models, reflecting its use of outcome-period data. However, classification performance remains similar across all three models, suggesting that it is possible to identify food-crisis areas with nearly the same accuracy even when predicting 12 months in advance.

Model Performance Relative to Benchmark Approaches

To assess the incremental predictive value of the ML models, we compare their performance with three benchmarks. Wherever possible, the baseline models use the same predictor information, training period, evaluation sample, and performance metrics as the corresponding ML models.

Persistence baseline. A persistence baseline assumes that the most recently observed IPC phase remains unchanged and uses it as the prediction for the future period. For the forecasting and nowcasting analyses, 67 of 1,170 observations in the holdout sample have no prior IPC assessment and therefore cannot receive a persistence-based prediction. For observations with at least one prior assessment, we use the most recent observed IPC phase as the predicted phase. We evaluate the persistence benchmark and the ML model on the same subset of observations to ensure comparability.

Classification-based baseline. We estimate an ordered probit model in which the outcome is the IPC phase and the explanatory variables are the same predictors used in the ML model. The ordered probit provides a conventional parametric benchmark that exploits the ordinal structure of the five IPC phases while imposing a much simpler functional form than the ML model. The model is trained and evaluated using the same temporal split as the corresponding ML specification.

Regression-based baseline. We estimate separate ordinary least squares (OLS) models for the population shares in Phase 5, Phase 4 or above, Phase 3 or above, and Phase 2 or above, using the same predictors as in the ML model. We then apply the same IPC 20% population-share rule used in the main analysis to derive the overall IPC phase. This provides a parametric benchmark for both the continuous population-share predictions and the resulting IPC phase classifications.

Panels A–C of Supplementary Table 1 report the baseline results, while Panels D and E report the corresponding ML results on comparable samples. The baseline models have lower overall phase-classification accuracy than the corresponding ML models. In general, the baselines achieve modestly higher precision, but at the cost of substantially lower sensitivity, particularly for forecasting and nowcasting. The OLS benchmark also performs worse than the corresponding ML models in predicting continuous population shares across all three settings. This is especially pronounced for forecasting and nowcasting, where the OLS models yield negative OOS R^2 values and therefore perform worse than a naive mean-prediction benchmark. Taken together, these results suggest that the ML models provide meaningful incremental predictive value for overall phase classification, crisis detection, and continuous population-share prediction.

Detecting More Severe Food Insecurity Phases

While the XGboost model showed promising results on generating Phase 3+ alerts, its skills in further separating the rarer Phases 4 and 5 from Phase 3 is more constrained. The confusion matrices in the left panels of Supplementary Fig. 4 show limited ability to distinguish these severe phases. Supplementary Fig. 5 assesses calibration of predicted cumulative IPC population shares against observed shares, not probabilities that an area experiences a food crisis. Predicted shares are positively associated with observed shares, but predictions for Phase 3+ and especially Phase 4+ are compressed relative to the observed distribution. In particular, the models substantially underpredict the largest observed Phase 4+ population shares. This pattern is consistent with the relatively small number of Phase 4/5 observations available for training (see Fig. 4).

To improve detection of severe cases, we perform an additional "rescue" step targeting observations classified as Phase 3 by the XGBoost model but belong to the rarer Phase 4 or 5. To do so, we train an XGBoost classifier specifically on observations in Phases 3–5, treating Phases 4/5 as the positive class and Phase 3 as the negative class. This more targeted setting helps the model to focus more on distinctive signatures within the Phase 3+ group. To account for class imbalance, we compare unweighted, square-root balanced, and fully balanced specifications. For forecasting and nowcasting, classification thresholds are selected using pre-2022 temporal cross-validation and performance is evaluated on the 1,170-observation 2022 temporal holdout sample. The contemporaneous model uses the fivefold cross-validation methodology to select threshold. Based on the F_2 criterion, which places greater weight on recall, we select the square-root balanced specification for forecasting and nowcasting and the fully balanced specification for the contemporaneous model. Results across weighting schemes are reported in Supplementary Fig. 6.

Under the selected specifications, the forecasting, nowcasting, and contemporaneous models identify approximately 75%, 84%, and 78% of true severe (Phase 4+) cases, respectively, while approximately 31%, 50%, and 14% of non-severe cases are incorrectly classified as severe. Supplementary Fig. 4 compares the original five-class predictions with rescue outputs that combine predicted Phases 4 and 5. The rescue procedure improves sensitivity to severe cases at the cost of additional false alarms and a reduction in overall five-class accuracy of approximately 13.7 percentage points. It should therefore be interpreted as a high-sensitivity screening approach for severe cases rather than as an improvement in overall five-class classification. These results underscore the continued importance of expert assessment particularly for severe cases that may depend on qualitative and context-specific information that is difficult to capture with statistical models [16]. Predictive performance is further constrained by the limited number of severe observations available for training.

Alternative Geospatial Modeling Strategies

In our main model, we include the latitude and longitude of each IPC area's centroid as predictors. To assess whether predictive performance depends on these coordinates, we conduct three robustness checks:

1. Exclude latitude and longitude, to assess whether model performance relies on explicit location information.
2. Replace them with k-nearest-neighbor spatial averages of each predictor across the five nearest IPC areas, to incorporate information from neighboring areas.
3. Replace them with spatial averages of each predictor across IPC areas within 200 km, to incorporate information from the broader local spatial context.

The two neighborhood-based specifications are intended to capture spatial dependence that may not be represented by an area's own predictors alone. In both specifications, the focal area is excluded because its information is already captured by its own predictor values.

The results are summarized in Supplementary Table 2. Excluding latitude and longitude produces little change in predictive performance, suggesting that geographic coordinates contribute little unique information once other covariates are included. Many predictors, including weather, agroecological characteristics, vegetation, market access, infrastructure, and conflict, are themselves spatially structured and may capture similar geographic heterogeneity.

The three specifications perform similarly in terms of classification metrics. However, for forecasting and nowcasting, the two spatial-average specifications yield lower OOS R^2 than the specification that simply excludes latitude and longitude. One possible explanation is that spatial averaging smooths away locally informative variation while introducing additional correlated predictors. Thus, although spatial dependence may remain important, these simple spatial-smoothing approaches do not improve predictive performance in our setting. We also note that these robustness checks do not constitute a comprehensive spatial modeling framework. Developing approaches that more explicitly capture adjacency, spatial dependence, and polygon characteristics remains an important direction for future research.

Exploratory Analysis of Generalization to New Geographic Areas

Our forecasting and nowcasting analyses are primarily designed to predict AFI in current and future periods for locations represented in the IPC data, rather than to extrapolate to entirely new geographic areas. Similarly, the contemporaneous analysis is intended to fill gaps in historical IPC assessments for areas that have been assessed by IPC at other points in time.

To explore the extent to which the models generalize to previously unseen geographic areas, we conduct two additional spatial validation exercises: leave-one-country-out analysis and five-fold area-level cross-validation. For forecasting and nowcasting, the leave-one-country-out analysis removes all observations from a given country from the training sample and uses observations from that country in the test period exclusively for out-of-sample evaluation. In the five-fold area-level cross-validation, areas represented in the test sample are randomly partitioned into five folds, with approximately 20% of areas assigned to each fold. For each fold, the corresponding areas are used for testing, while all observations from those areas are excluded from

the training sample. Thus, the models are evaluated on future observations from geographic units that have never appeared in the training data. For the contemporaneous analysis, the same geographic grouping principles are applied to both exercises without a temporal split, with countries or areas assigned to mutually exclusive training and test folds.

Supplementary Table 3 shows that population-share prediction deteriorates more noticeably when forecasting and nowcasting are extended to previously unseen geographic areas. Phase 3+ population-share R^2 declines from 0.249 and 0.253 in the standard forecasting and nowcasting evaluations to 0.165 and 0.170 under area-level cross-validation. Under leave-one-country-out validation, R^2 falls to -0.014 for forecasting and -0.007 for nowcasting, indicating that predicting continuous population shares in entirely unseen countries is substantially more difficult. In contrast, phase-classification performance remains relatively strong under both geographic validation exercises, suggesting that classification transfers more robustly across locations than continuous population-share prediction. For contemporaneous prediction, population-share performance is stronger under area-level cross-validation, but declines substantially under leave-one-country-out validation. Taken together, these results suggest that the models generalize relatively well to previously unseen areas within countries represented in the training data, particularly for phase classification and contemporaneous estimation, whereas extrapolation to entirely unseen countries remains considerably more challenging.

Discussion

In this paper, we predict food crises—as defined by IPC measures of AFI—using passively collected big data. The IPC is widely regarded as the gold standard for assessing AFI and is extensively used by humanitarian and development organizations. However, collecting IPC data is often costly and logistically challenging, particularly in fragile and conflict-affected settings. Recent advances in remote sensing have enabled access to large-scale, passively collected datasets that correlate with AFI risk factors. These developments make it possible to monitor relevant risk factors in near real-time and predict food crises more efficiently, at lower cost, and in regions that are otherwise hard to reach.

Our forecasting model correctly identified 94.1% of food crisis areas in the test set with a 12-month lead time. Among the areas flagged as at risk, 77.5% were confirmed as true crisis areas. These results demonstrate the feasibility of using real-time monitoring of risk factors to forecast food crises and highlight the potential of our predictive model to support EWS and facilitate timely humanitarian responses.

An important policy implication of this study is that earlier identification of emerging food insecurity can create opportunities for anticipatory action that may help prevent or attenuate deterioration into IPC Phase 3 or above. The 12-month forecasting horizon provides time for governments, humanitarian organizations, and development partners to mobilize financing, pre-position food and therapeutic nutrition supplies, and expand shock-responsive social protection before household

food-consumption gaps become severe. Because the model also predicts the proportion of the population in food crisis, its results may provide an additional indication of where needs are likely to be greatest and of their potential scale. Given the model's more limited performance in predicting population shares, however, these estimates are better suited for screening and prioritization than for precisely estimating the number of people in need of assistance.

Several additional steps would be needed before the model could be used operationally. Although we employ forward-looking time-series validation during model development, the final out-of-sample evaluation covers only one calendar year, 2022. Evaluating the model over additional years and across a wider range of crisis conditions would provide stronger evidence of temporal generalizability. Further improvements in predictive performance, model updating, data timeliness, and integration with existing early-warning workflows would also be necessary before operational deployment.

Feature-importance measures should also be interpreted cautiously. They identify variables that are predictive of food crises but do not establish that changing those variables through a particular policy would causally prevent a crisis. The model should therefore be viewed as an early-warning, screening, and resource-prioritization tool that complements, rather than replaces, IPC assessments, local contextual analysis, and expert judgment. Its predictions can indicate where and when further assessment and anticipatory action may be warranted, while the choice of intervention should be guided by contextual evidence on the underlying drivers of food insecurity.

Methods

The Gradient Boosting Method

We compile a dataset of 5,575 observations covering 1,198 subnational areas in 29 countries from 2017 to 2022. The unit of analysis is the subnational area and month for which IPC AFI measures are available. The outcome variables include both the overall IPC Phase and the population shares across each IPC Phase. Among the predictor variables, we incorporate key drivers of contemporary food crises such as conflict, weather shocks, and economic shocks.

We employ a machine learning algorithm based on Gradient Boosting [17], a widely used ensemble learning technique. Gradient Boosting builds a strong predictive model by combining multiple weak learners, typically decision trees, through iterative training that corrects errors from previous rounds. We choose this method for its strong performance in classification tasks, robustness to outliers, compatibility with diverse data types, and built-in regularization features that help prevent overfitting. Specifically, we use the XGBoost (eXtreme Gradient Boosting) implementation developed by [18], which is well-regarded for its computational efficiency and high accuracy.

We further implement an ensemble modeling approach that integrates four separate predictive models, as illustrated in Fig. 3. This structure mirrors the classification logic outlined in the FAO IPC manual [3]. Each model focuses on predicting the percentage of the population in a specific IPC phase threshold: Phase 2 or higher, Phase 3 or higher, Phase 4 or higher, and Phase 5. These models collectively produce predicted population shares falling into each severity category for every subnational area.

The overall IPC phase classification is then determined using a threshold-based rule: if at least 20% of the population is predicted to be in Phase 5, the area is classified as IPC Phase 5. If fewer than 20% are predicted in Phase 5 but at least 20% are predicted in Phase 4 or higher, the area is classified as Phase 4, and so on, down to Phase 1. This procedure reflects the operational approach used in IPC assessments. Finally, the model-generated IPC phase labels are compared to the actual IPC classifications to evaluate predictive accuracy and performance.

Performance Evaluation

We assess model performance using out-of-sample R^2 when the outcome variables represent the percentage of the population in food crisis (Phase 3 or above). For five-class IPC phase classification, accuracy is defined as the proportion of evaluated observations for which the predicted phase matches the observed phase. We separately evaluate binary food-crisis classification, defining IPC Phase 3 or above as the positive class. Sensitivity is the proportion of actual crisis observations correctly identified as crises, while precision is the proportion of predicted crisis observations that are actual crises. Sensitivity therefore captures the model's ability to identify areas experiencing food crises, whereas precision reflects the reliability of crisis alerts and is particularly relevant for targeting humanitarian response and allocating limited resources.

Hyperparameter Selection

Forecasting and Nowcasting Models. Hyperparameter tuning was performed for forecasting, following [8]. Nowcasting and contemporaneous models were subsequently evaluated with fixed hyperparameters, without separate hyperparameter searches. For forecasting and nowcasting, January 2017 to December 2021 data were used for training, and January to December 2022 data were reserved for final model evaluation.

Forecasting hyperparameters were selected by grid search with four expanding-window temporal validation folds, as illustrated in Supplementary Fig. 3. The first fold used 2017 for training and 2018 for validation. The training window then expanded annually, with validation in 2019, 2020 and 2021. The 2022 holdout was not used for hyperparameter selection.

Candidate configurations were ranked by their mean validation score across the four folds, combining accuracy, precision, sensitivity and Phase 3+ R^2. Sensitivity received a weight of five; each other metric received a weight of one. The search grid, two-stage procedure and effective parameter values are reported in the Supplementary Notes.

In the first tuning stage, we selected hyperparameters for the Phase 3+ cumulative target. In the second stage, we held this component fixed while selecting hyperparameters for the Phase 2+, Phase 4+, and Phase 5 cumulative targets.

The optimized hyperparameters include maximum tree depth, number of estimators, learning rate, subsample ratio, column sample by tree, gamma, minimum child weight, and scale-positive weight.

Building on the hyperparameter tuning and holdout validation methods described above, we simulate the forecasting scenario using predictors lagged by 12 months. In

contrast, since nowcasting incorporates both current and lagged predictors, we develop a cascading nowcasting model to complement our forecasting approach. This two-layer framework operates as follows: the first layer generates forecasts using only 12-month lagged predictors (e.g., conflict, weather, agricultural, and economic indicators); the second layer refines these predictions by integrating current and recent (within 12 months) features. These features capture updated information and residual signals that emerge after the initial forecast. This design allows us to exploit both the structural robustness of the forecasting model and the immediacy of newly available data at the time of prediction.

Both layers of the nowcasting model use the forecasting-selected hyperparameters, without additional tuning. The cascading design also offers practical advantages by aligning closely with potential operational workflows. As emphasized in the IPC handbook [2], timely updates are essential. Our approach enables policymakers and analysts to use an earlier forecast as a baseline. When new, contemporaneous data becomes available, only the computationally lightweight second layer needs to be executed, using updated features. This allows for a refined, near-real-time nowcast without rerunning the more resource-intensive full forecasting model—minimizing computational cost while maximizing responsiveness.

Together, the integrated forecasting and two-layer nowcasting framework fulfill a dual role critical to effective food security monitoring and response. The forecasting component facilitates EWS activation by projecting emerging risks, while the nowcasting component provides timely updates that help identify geographic areas requiring immediate, targeted intervention based on the latest observed conditions.

Contemporaneous Prediction Model. Contemporaneous performance was evaluated using random five-fold, row-level cross-validation over all 5,575 observations from January 2017 to December 2022 [11]. Observations were assigned to five equal folds using a fixed random seed of 0. Each iteration trained on four folds (80%; 4,460 observations) and predicted the remaining fold (20%; 1,115 observations). Hyperparameters remained fixed throughout, with no separate validation split or hyperparameter search. Each observation received one out-of-fold prediction, and reported metrics were calculated from the pooled predictions rather than averaged across folds. This random-fold evaluation differs from the 2022 temporal holdout used for forecasting and nowcasting.

Data

Outcome Variables

Our outcome variables are based on historical IPC reports released by IPC Global Partnership, covering the period from January 2017 to December 2022. The IPC framework categorizes the severity of AFI into five phases [3]:

- Phase 1: Minimal — Food security conditions are stable, and basic food needs are met.

- Phase 2: Stressed — Households have minimally adequate food consumption but cannot afford some essential non-food expenditures without engaging in stress-coping strategies.
- Phase 3: Crisis — Households have food-consumption gaps reflected in high or above-usual acute malnutrition, or barely meet minimum food needs by depleting essential livelihood assets or adopting crisis-coping strategies.
- Phase 4: Emergency — Households face large gaps in food consumption, very high acute malnutrition, or excess mortality.
- Phase 5: Catastrophe/Famine — Households experience an extreme lack of food and other basic needs where starvation, death, destitution, and extremely critical acute malnutrition levels are evident.

In addition to phase-specific classifications, the IPC provides estimates of the population falling into each phase within a given geographic area. According to IPC Global Partners, an area is classified as being in a food crisis when at least 20% of its population is in IPC Phase 3 or above. This threshold serves as a key operational indicator for assessing the severity of food insecurity and guiding the prioritization of humanitarian responses.

To ensure data quality, we exclude all observations without valid IPC phase identification and cross-reference the IPC database to confirm these are not attributable to data entry errors. The resulting dataset includes 5,575 observations across 1,198 subnational areas in 29 countries from 2017 to 2022. Fig. 4 Panel A shows the distribution of IPC phases: Phase 3 accounts for 52.4% of observations, followed by Phase 2 (35.3%), Phase 4 (8.7%), Phase 1 (3.4%), and Phase 5 (0.3%). Panel B of Fig. 4 displays the distribution over time, which mirrors the overall pattern—Phases 1 and 5 occur infrequently, while the majority of observations fall into Phase 3.

Predictor Variables

Our selection of predictor variables is guided by potential risk factors and key drivers of food insecurity identified in the literature (e.g., [12]), and includes several variables currently used by the IPC system for monitoring food crises. All variables are publicly available, and their sources are listed in Supplementary Table 4.

To integrate secondary geospatial raster data with our analytical units, we apply a systematic geoprocessing workflow. We begin by defining and validating the IPC polygonal boundaries for our areas of interest to ensure geometric integrity. All spatial datasets are standardized to the WGS 84 geographic coordinate reference system (EPSG:4326), with longitude and latitude expressed in degrees. Vector polygons or time-series raster imagery are transformed as needed for spatial overlays. Each dated raster file—representing specific environmental or climatic variables—undergoes a quality assessment to identify no-data values and to confirm suitability for analysis. We use geodesic area calculations for the raster–polygon overlap assessment. Given the variable spatial resolutions of the raster datasets, we include a pixel in a polygon's aggregation if more than 50% of its area falls within that polygon.

We extract the data using ArcPy, the ArcGIS Python environment. For each IPC polygon and raster date, we isolate pixel values within the polygon boundaries and

compute the spatial average for each variable, excluding no-data values to avoid bias. We apply this process across all variables and time periods, compiling the results into a structured spatio-temporal dataset indexed by polygon and month. We then perform final checks to ensure the completeness and validity of the data.

The resulting predictor dataset includes 172 variables spanning physical geography, food prices, weather conditions, other economic indicators, agricultural and land productivity, and conflict data.

Physical Geography. Elevation is a critical factor in climate modeling, affecting temperature, precipitation patterns, and ecosystem distribution. We use elevation data from the ECMWF ERA5-Land reanalysis [19], recorded as elevation in the predictor table.

Ruggedness, defined as the degree of elevation variability within a given area, captures terrain complexity and unevenness. It plays an important role in shaping economic geography, particularly by affecting the feasibility and cost of transportation, trade, and productive activities. Although its impact can vary with technological development and economic structure, high ruggedness is frequently associated with limited accessibility and economic underdevelopment. We extract ruggedness values at the pixel level from the dataset developed by [20].

Distance to rivers measures the proximity of a given area to the nearest major river system, a factor that significantly affects transportation, agricultural productivity, and economic development. River access facilitates irrigation, logistics, and trade, and has long played a strategic role in shaping settlement patterns and economic outcomes. We calculate this variable using ArcGIS by sourcing river shapefiles from the World Bank and the global hydrography dataset by [21]. For each IPC area, we compute the centroid distance to the nearest river, ensuring accurate spatial representation for subsequent analysis.

Food Prices. Food price data are obtained from two primary sources: the World Food Programme (WFP) and the Food and Agriculture Organization (FAO). From WFP, we utilize two monthly country-level indicators—Percentage of Price Change and Spatial Price Dispersion—to capture market shocks and assess the extent of market fragmentation. Both indicators are based on the prices of a food basket comprising four staple commodities: rice, wheat, maize, and vegetable oil [22].

The Percentage of Price Change measures the month-on-month change in food prices relative to the preceding month, expressed as a percentage. Spatial Price Dispersion, on the other hand, is measured as the standard deviation of local market prices within a country, capturing geographic variability in food access. To capture temporal dynamics, we construct four time-series features from each of these two indicators: the 12-month lag, 12-month moving average, 12-month lagged moving average, and the 12-month temporal standard deviation.

To further quantify interannual trends and volatility in staple food prices, we use the FAO-GIEWS Food Price Monitoring and Analysis (FPMA) database [23]. This source provides monthly wholesale and retail price data for individual commodities across markets within a country or territory. Each recorded price is divided by its

commodity-specific reference price to form a dimensionless price index. We do not apply a price deflator or any other inflation adjustment. These indexed prices are averaged at the market-month level across all commodities available in that market-month, giving equal weight to each commodity. The data retain the market's geographic coordinates and name. Markets are matched to the nearest administrative unit using great-circle distance, and any market located more than 100 km away from the nearest admin unit centroid is excluded. For the remaining markets in the dataset, we compute the 12-month lag, 12-month moving average, 12-month lagged moving average, and the 12-month temporal standard deviation.

We also construct the variable Distance to the Nearest Market using FAO FPMA price records [23] to provide additional insights into market accessibility for both agricultural producers and consumers. Using the detailed geographic coordinates provided in the FPMA dataset, we calculate the distance from each polygon centroid to the nearest market using the Haversine formula [24].

Weather Conditions. We use the 2m temperature variable from ERA5-Land monthly averaged data [25], representing air temperature at 2 meters above the surface. ERA5-Land has an approximately 9 km native resolution; the Climate Data Store distributes these data on a regular $0.1^\circ \times 0.1^\circ$ latitude–longitude grid. We compute the spatial mean for each polygon and month. The model inputs distinguish monthly updated 12-month rolling means from 12-month-lagged values.

We also use the ERA5-Land "Surface solar radiation downwards" (SSRD) variable [19] to characterize solar energy availability. For each polygon, we compute the spatial mean of daily total SSRD and then average these daily totals by month. The model inputs include monthly updated 12-month rolling means and 12-month-lagged values.

Precipitation data include the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) [26], which provides daily and monthly estimates over land between 50°S and 50°N at 0.05° (approximately 5 km) spatial resolution. We spatially average monthly-aggregated CHIRPS rainfall within each polygon. The resulting inputs include monthly updated 12-month rolling means and 12-month-lagged values. ERA5-Land precipitation is included separately, using the same rolling-mean and lag notation [19].

Temperature and CHIRPS rainfall z-scores use a fixed 2002–2022 reference period within each polygon. The contemporaneous fields are annual values, repeated across observations within the same polygon and calendar year. The forecasting inputs instead contain the separate lagged z-score fields which vary within calendar years. These lagged series are distinguished from the annual z-score fields rather than treating all z-score predictors as having the same update frequency. The second layer of nowcasting uses the current rolling-mean and annual z-score fields; its first layer uses the forecasting inputs. Contemporaneous models use both current and lagged inputs.

Other Economic Conditions. To contextualize the socioeconomic environment relevant to food security, we incorporate five country-level indicators sourced and distributed from the World Bank Group (WBG). These are the Consumer Price Index (CPI), GDP per capita (constant 2010 US$), the Liner shipping connectivity index

(LSCI) infrastructure score, the Gini Index, and the Control of Corruption indicator from the Worldwide Governance Indicators (WGI) [27, 28].

- CPI: We use the consumer price index level (2010 = 100; indicator code `FP.CPI.TOTL`), which measures consumer prices relative to the reference year. We retain the index level without converting it to an annual percentage change.
- GDP per capita: Represents the average economic output per person, offering insight into national wealth and economic productivity.
- Infrastructure Index: We use the Liner Shipping Connectivity Index (LSCI; indicator code `IS.SHP.GCNW.XQ`), which measures a country's integration into global liner shipping networks. Higher values indicate stronger maritime transport connectivity and better access to international trade networks, reflecting the availability and quality of regular shipping services and port connections.
- Gini Index: Quantifies income or consumption expenditure inequality, ranging from 0 (perfect equality) to 100 (perfect inequality), reflecting disparities within the population.
- Control of Corruption Indicator: Assesses perceptions of public power misuse for private gain, covering both minor and large-scale corruption as well as state capture by elites. It is expressed as a percentile rank (0–100), with higher values indicating stronger control over corruption.

These country-level indicators enter our country-year dataset to characterize economic conditions, consumer-price levels, infrastructural capacity, income distribution, and governance quality. We also include their 12-month lagged values as additional predictors in the analysis.

Night-time light intensity provides a proxy for economic activity, urbanization, and human presence. We use monthly night-time light composites from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band [29]. These satellite observations provide information on economic activity where reliable monthly economic indicators are scarce.

We calculate the 12-month rolling mean and standard deviation of monthly VIIRS night-time light intensity as measures of development and temporal variability. We also include the 12-month lag of the rolling mean as a lagged predictor.

Travel time to cities is a key indicator of access to economic resources and urban centers. Sourced from Weiss (2018) [30], it maps global travel times to cities in 2015 at a one-square-kilometer resolution. We compute travel time by extracting and averaging values from geographic polygons. This metric captures access to urban opportunities and services, accounting for distance, transport infrastructure, and city distribution. Limited connectivity can hinder livelihoods and development, underscoring the importance of infrastructure for economic growth.

Agriculture and Land Productivity. We use the Global OCO-2 SIF (Solar-Induced Fluorescence) based Gross Primary Production (GOSIF-GPP) and the Enhanced Vegetation Index (EVI) to monitor land productivity dynamics. GOSIF-GPP estimates the total carbon dioxide converted to organic carbon via photosynthesis and serves as a high-resolution proxy for terrestrial productivity. It integrates data

from: (1) OCO-2 SIF, a direct indicator of photosynthetic activity, (2) MODIS EVI, which reflects vegetation health, and (3) MERRA-2 meteorological data, which accounts for climatic influences.

The GOSIF-GPP dataset offers monthly estimates at a 0.05° spatial resolution and is highly correlated with ground-based GPP measurements, providing insights into ecosystem productivity and health [31]. For each geographic unit, we calculate the spatial mean and construct: (1) a 12-month rolling average to capture long-term trends, (2) the 12-month standard deviation to assess seasonal variability and crop stress, and (3) a 12-month lag of the rolling average to serve as a predictive indicator of delayed productivity responses.

EVI is a satellite-derived vegetation index designed to quantify greenness and plant health with improved sensitivity over traditional indices, particularly in densely vegetated regions [32]. Derived from MODIS sensors at a 250m resolution, it enables detailed analysis of vegetation dynamics and responses to environmental changes.

For our analysis, we extract 16-day EVI raster files based on predefined geographic shapes and integrate them into our dataset. From these, we compute: (1) a 12-month moving average to identify long-term vegetation trends, (2) the 12-month standard deviation to capture seasonal fluctuations, and (3) a 12-month lag of the rolling average to reflect lagged vegetation conditions.

Soil moisture is a key indicator of drought conditions and plays a critical role in agricultural planning and ecosystem health. We incorporate monthly average soil moisture data from ERA5-Land [19]. We compute a 12-month rolling mean and standard deviation to capture long-term trends and periods of high variability that may affect plant growth and water availability. In addition, we include a 12-month lag of the rolling average to serve as a lagged predictive indicator of soil conditions.

We also integrate soil property data from the International Soil Reference and Information Centre (ISRIC) SoilGrids, a global digital soil mapping platform that employs advanced machine learning techniques [33]. SoilGrids provides global estimates of soil properties at six standard depth intervals with 250-meter spatial resolution. Available attributes include pH, soil organic carbon content, bulk density, coarse fragment content, sand, silt, and clay composition, cation exchange capacity (CEC), total nitrogen, soil organic carbon density, and carbon stock. For our analysis, we extract the spatial mean of each property at the near-surface depth (0–5 cm) using raster data.

To characterize agricultural land cover, we use cropland and rangeland masks from the Anomaly Hotspots of Agricultural Production (ASAP) system [34]. ASAP is developed and maintained by the European Commission's Joint Research Centre (JRC). These masks provide a foundational layer for agricultural and food security analyses. We extract the data using predefined geographic polygons and integrate it into our dataset as two separate variables: the percentage of land area classified as cropland and the percentage classified as rangeland.

Furthermore, we incorporate Agro-Ecological Zones (AEZ) from [35], which classify geographic regions based on ecological, agricultural, and soil characteristics, as well as maize and cereal cultivation patterns. AEZs are critical for capturing regional heterogeneity that influences agricultural production and food security outcomes. We

extract AEZ data using predefined geographic polygons and generate dummy variables to represent the specific zones.

In the countries included in our study, the major AEZs—ranked by land area—are as follows: Tropical Maize Mixed (TMS), prevalent in Zambia, Angola, and parts of Tanzania; Tropical Monsoon Bimodal (TMB), covering Kenya, coastal Tanzania, and Haiti; Tropical Monsoon Rice/Irrigated (TMR/TMI), found in Bangladesh; Tropical Highland Mixed (THR), present in Ethiopia and Guatemala; Hot Arid Irrigated (HAI), spanning parts of Sudan and Pakistan; Hot Semi-Arid Winter-Cereal (HSW), covering Afghanistan and western Pakistan; Tropical Rainforest Bimodal (TRB), occurring in the Democratic Republic of Congo and eastern Andean Ecuador; Mediterranean Winter Cereal (MED), located around Lebanon and Palestine; and Temperate Southern Hemisphere (TSH), found in South Africa and Lesotho.

Conflict. Conflict is a major driver of instability and insecurity. To measure its impact, we use two key metrics:

- Conflict Incidence: Monthly counts of violent events, recorded separately for each event type.
- Conflict Severity: Monthly fatality counts, recorded separately for the same event types.

These metrics are derived from the Armed Conflict Location and Event Data (ACLED) dataset [36], which provides precise geographic coordinates for conflict events. Using these coordinates, we calculate distances from conflict points to the polygon centroids of sampled areas.

To generate the inverse distance-weighted frequencies, we multiply the total number of conflicts or fatalities at each location by the inverse of its distance to the centroid. This produces the monthly conflict and fatality frequencies, weighted by proximity. We apply this calculation using the 5 and 10 nearest conflict locations, representing different spatial scales of influence. Smaller numbers indicate a stronger weighting on nearby events.

We analyze various types of conflict events, including:

- Battles: Counts and fatalities from direct combat, providing insights into the intensity of armed engagements.
- Explosions: Counts and deaths caused by explosive devices, reflecting the human cost of such tactics.
- Violence Against Civilians: Counts and fatalities from targeted attacks on non-combatant populations, highlighting the toll on civilians.

For each event type, the inputs retain monthly counts and the corresponding distance-weighted variants, identified by the suffixes `_w5` and `_w10`. The nowcasting inputs additionally include monthly updated 12-month rolling means, denoted by `_m12`; forecasting uses 12-month-lagged fields, denoted by `_l12`. For example, `event_count_battles_w5_m12` is a rolling-mean input, whereas `event_count_battles_w5_l12` is a lagged input. The contemporaneous models retain both current and lagged conflict predictors.

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Competing interests

The authors have no competing interests to declare.

Data availability statement

All data necessary for replication of the results presented in this paper will be available in a public GitHub repository upon publication.

Code availability statement

The code necessary for replication of the results presented in this paper will be available in a public GitHub repository upon publication.

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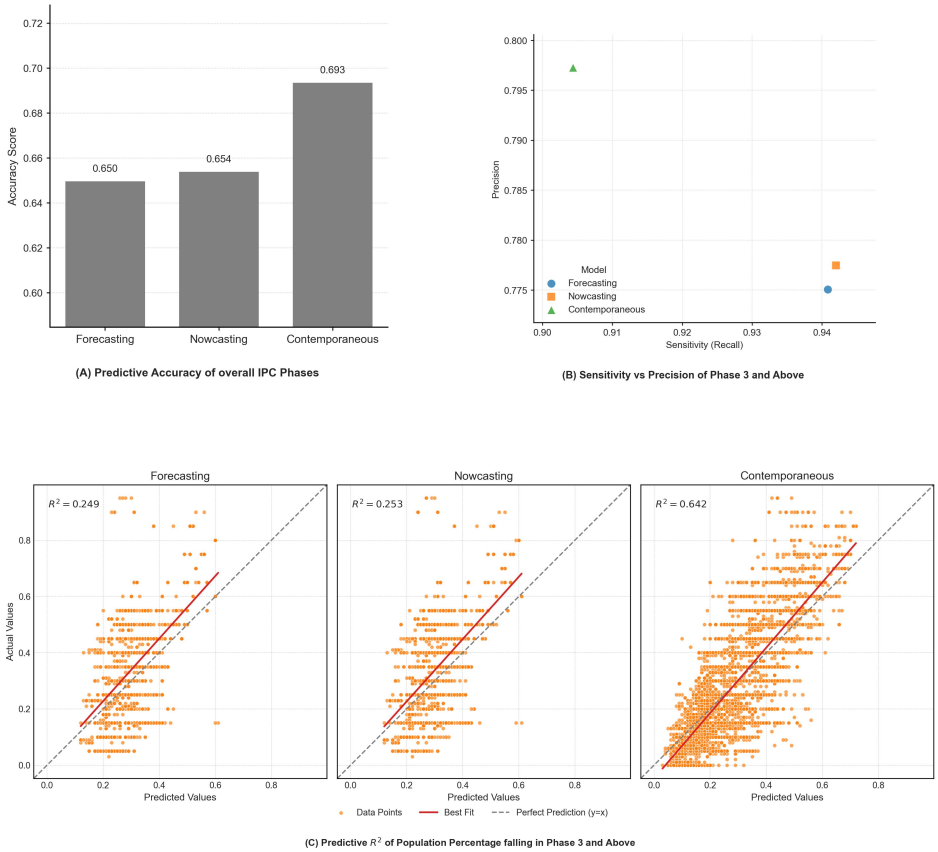
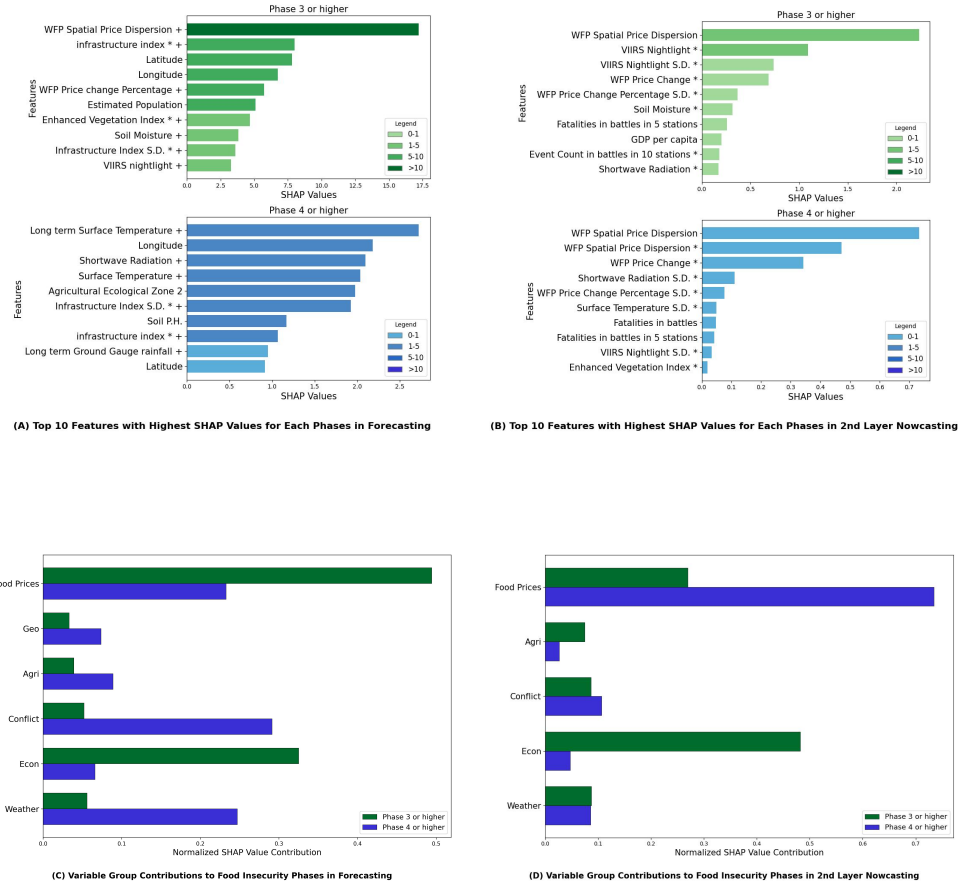


Fig. 1 Performance Metrics Across Forecasting, Nowcasting, and Contemporaneous Models

Notes: Panel A illustrates the model's accuracy in predicting the overall IPC (Integrated Food Security Phase Classification) phase, measured as the proportion of correctly predicted phases relative to the total number of actual cases. Panel B presents the sensitivity and precision of the model's alert classifications. An "alert" is defined as cases where the overall IPC phase is 3 or higher, signaling the need for urgent action. Sensitivity is the proportion of correctly identified alert cases out of all actual alert cases, while precision is the proportion of true alerts among all predicted alerts. Panel C displays the out-of-sample R^2 for the percentage of the population classified in Phase 3 or worse. It plots the predicted values against the actual observed values. In all three panels, forecasting and nowcasting use the 2022 temporal holdout (n = 1,170 each), while contemporaneous results use pooled random five-fold, row-level out-of-fold predictions (n = 5,575; seed 0). Evaluation populations and validation protocols therefore differ across models.



(A) Top 10 Features with Highest SHAP Values for Each Phases in Forecasting

(B) Top 10 Features with Highest SHAP Values for Each Phases in 2nd Layer Nowcasting

(C) Variable Group Contributions to Food Insecurity Phases in Forecasting

(D) Variable Group Contributions to Food Insecurity Phases in 2nd Layer Nowcasting

Fig. 2 Bar Chart of SHAP Value of Forecasting and Nowcasting Features

Notes: These four panels display SHAP values, which quantify the contribution of individual variables or variable groups to food insecurity predictions. Panel A presents the SHAP values of the top individual variables in the forecasting model for Phase 3+ (top figure) and Phase 4+ (bottom figure). Panel B shows the SHAP values of the top variables in Layer 2 of the nowcasting model, again for Phase 3+ (top) and Phase 4+ (bottom). Panel C displays SHAP values aggregated by variable group in the forecasting model, with Phase 3+ results shown in green and Phase 4+ in blue. Variables are grouped into: food prices, physical geography (Geo), agriculture and land productivity (Agri), conflict, economic conditions (Econ), and weather conditions. Panel D presents the SHAP values of the same variable groups, but for Layer 2 of the nowcasting model. In the graph, a variable name ending with "+" indicates it is lagged by 12 months, while a name ending with "*" denotes a 12-month moving average.

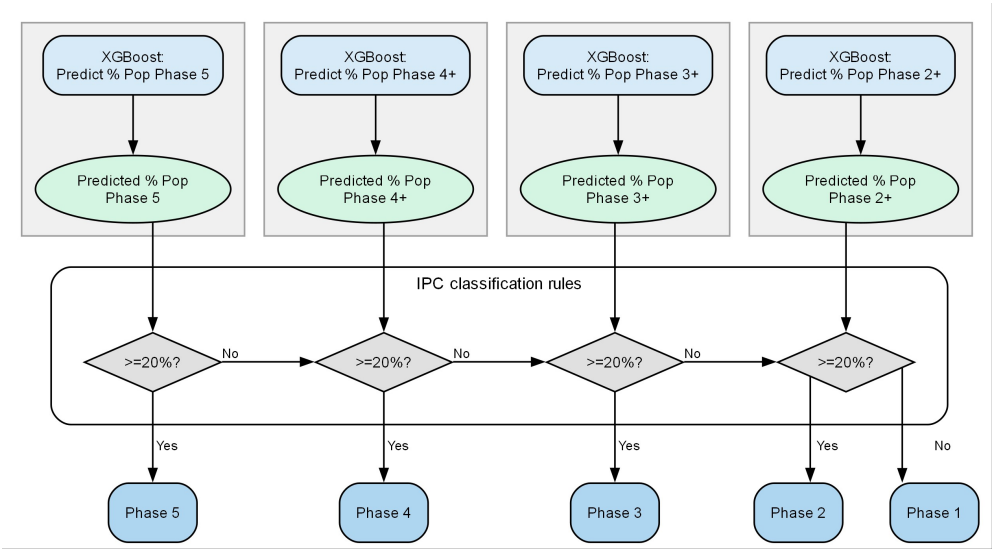


Fig. 3 Ensemble Model Procedure Overview
Notes: This flowchart illustrates the estimation process for IPC outcome variables, namely the overall IPC phase and the percentage of the population falling into each phase. The structure follows the classification logic outlined in the FAO IPC Manual [3]. We employ four separate XGBoost regression models to estimate the share of the population falling into: (i) Phase 2 or higher, (ii) Phase 3 or higher, (iii) Phase 4 or higher, and (iv) Phase 5. These outputs are combined to generate predicted population distributions across IPC phases. The overall IPC phase classification is then assigned based on a threshold rule: if at least 20% of the population is predicted to be in Phase 5, the area is classified as IPC Phase 5; if less than 20% is in Phase 5 but at least 20% is in Phase 4 or higher, it is classified as Phase 4, and so on, down to Phase 1.

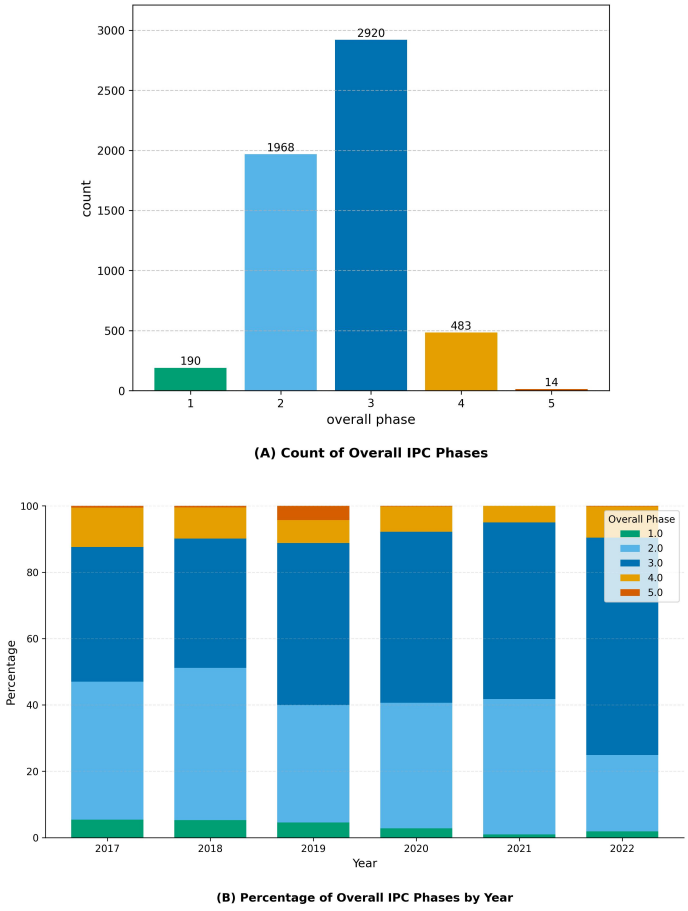


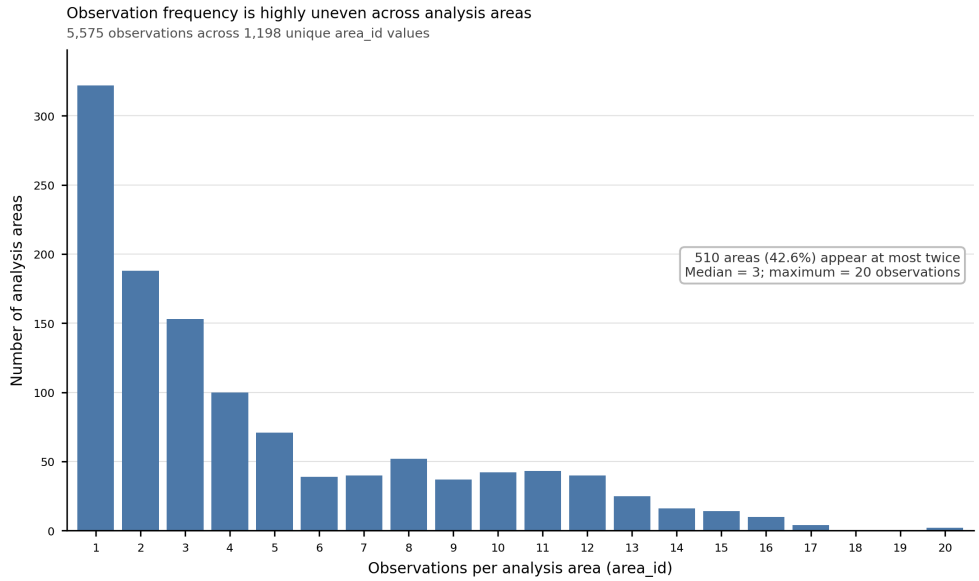
Fig. 4 Distribution of Overall Phase
Notes: Panel A displays the number of instances in each IPC AFI phase from 2017 to 2022. Phase 1: Minimal, Phase 2: Stressed, Phase 3: Crisis, Phase 4: Emergency, Phase 5: Famine. Panel B presents the percentage distribution of cases across IPC AFI phases for each year from 2017 to 2022. These descriptive distributions use the overall phase labels recorded in the source data. Contemporaneous model evaluations instead use phase labels reconstructed from the observed cumulative population shares using the 20% threshold rule; class counts can therefore differ even when the same observations are included.

Table 1 Performance of Forecasting, Nowcasting, and Contemporaneous Models

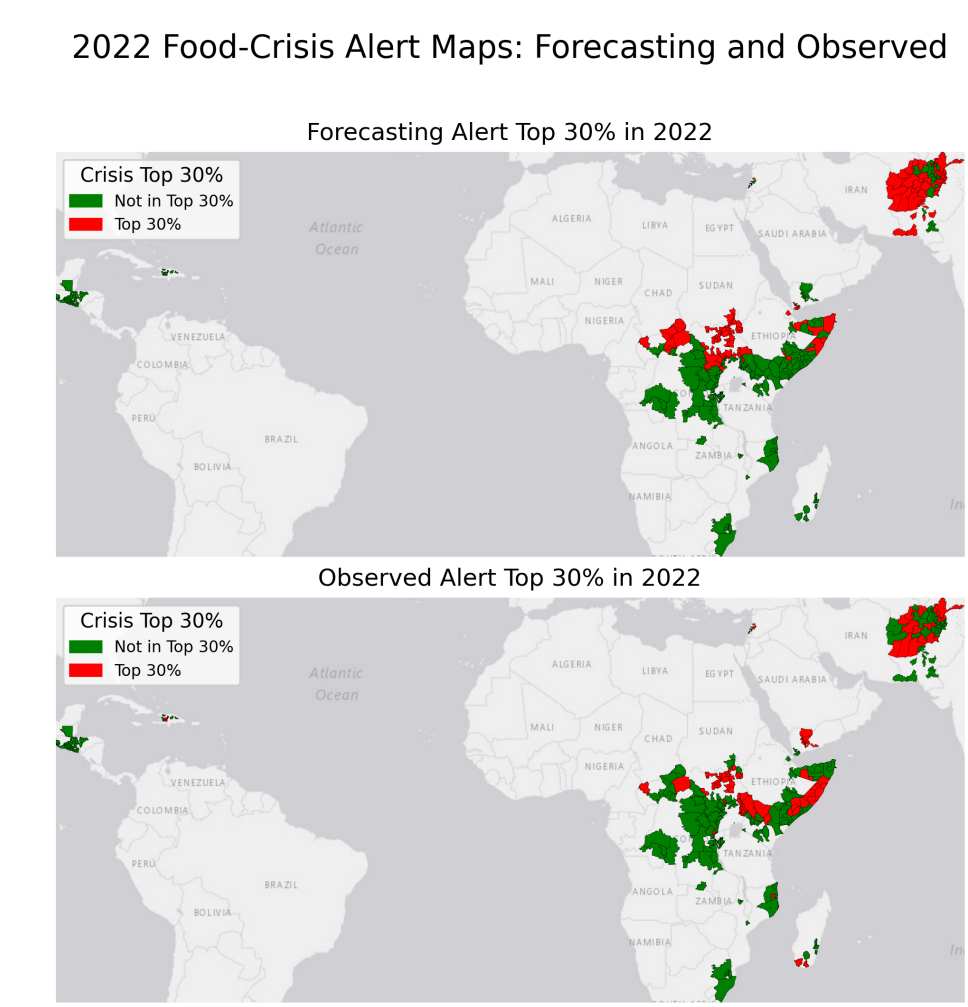
	Performance Metrics			
	(1) Accuracy	**(2) Sensitivity**	**(3) Precision**	**(4) R^2**
Panel A: Forecasting				
All Observations	0.650	0.941	0.775	0.249
Observations with Changed Phase	0.593	0.990	0.702	–
Panel B: Nowcasting				
All Observations	0.654	0.942	0.777	0.253
Observations with Changed Phase	0.571	0.964	0.733	–
Panel C: Contemporaneous				
All Observations	0.693	0.904	0.797	0.642

Notes: This table summarizes performance metrics for the forecasting, nowcasting, and contemporaneous models. Forecasting and nowcasting are trained on IPC data from 2017 to 2021 (n = 4,405) and evaluated on the 2022 temporal holdout (n = 1,170). Contemporaneous performance is computed from pooled random five-fold, row-level out-of-fold predictions for all 5,575 observations from 2017 to 2022 (seed 0; 4,460 training and 1,115 test observations per fold). Accuracy is defined as the proportion of correctly identified IPC phases relative to the total number of actual cases. Sensitivity and precision assess the models' ability to predict alerts, where an alert is defined as an overall IPC phase of 3 or higher, signaling the need for immediate action. Sensitivity is the proportion of true alert cases correctly identified, while precision is the proportion of true alerts among all predicted alerts. The out-of-sample R^2 evaluates the model's performance in predicting the percentage of the population classified as Phase 3 or above.
For forecasting and nowcasting, performance is reported for two subsets of the final evaluation sample: "All Observations," which includes all observations in 2022 holdout, and "Observations with Changed Phases," which includes only those observations where the overall IPC phase differs from that in the most recent prior assessment available in the training data.

Supplementary Figures

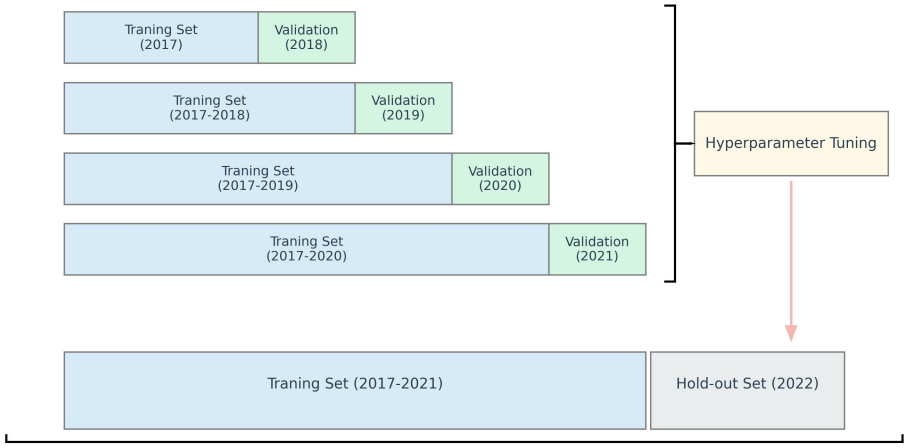


Supplementary Fig. 1 Uneven Observation Frequency Across Analysis Areas
Notes: The histogram shows the number of observations available for each analysis area. The horizontal axis indicates the number of observations associated with an individual area, and the vertical axis indicates the number of unique analysis areas with that observation frequency. The sample contains 5,575 observations from 1,198 unique analysis areas. Observation frequency is strongly unbalanced: 510 areas (42.6%) appear no more than twice, the median area appears three times, and the maximum number of observations for a single area is 20.

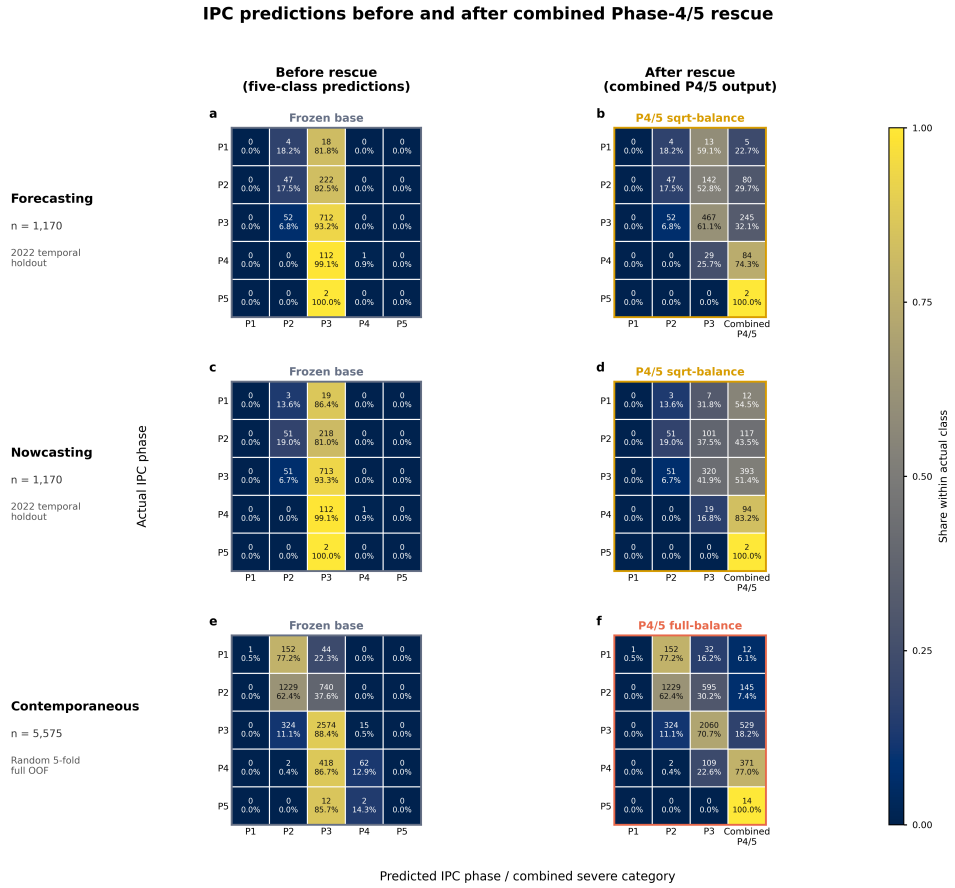


Supplementary Fig. 2 Top 30% Predicted and Observed Food-Insecure Areas by Population Share in Food Crisis

Notes: The maps compare the spatial distribution of analysis areas in the highest 30% of the population share experiencing food crisis. Red areas in the upper panel indicate locations predicted, 12 months in advance, to fall within the top 30%, while red areas in the lower panel indicate locations in the top 30% based on observed outcomes. The remaining evaluated areas are shown in green. Using the same threshold facilitates comparison of the geographic concentration and correspondence between predicted and observed high-risk areas. Gray background regions and place labels are provided only for geographic reference and do not indicate the availability of evaluation observations.

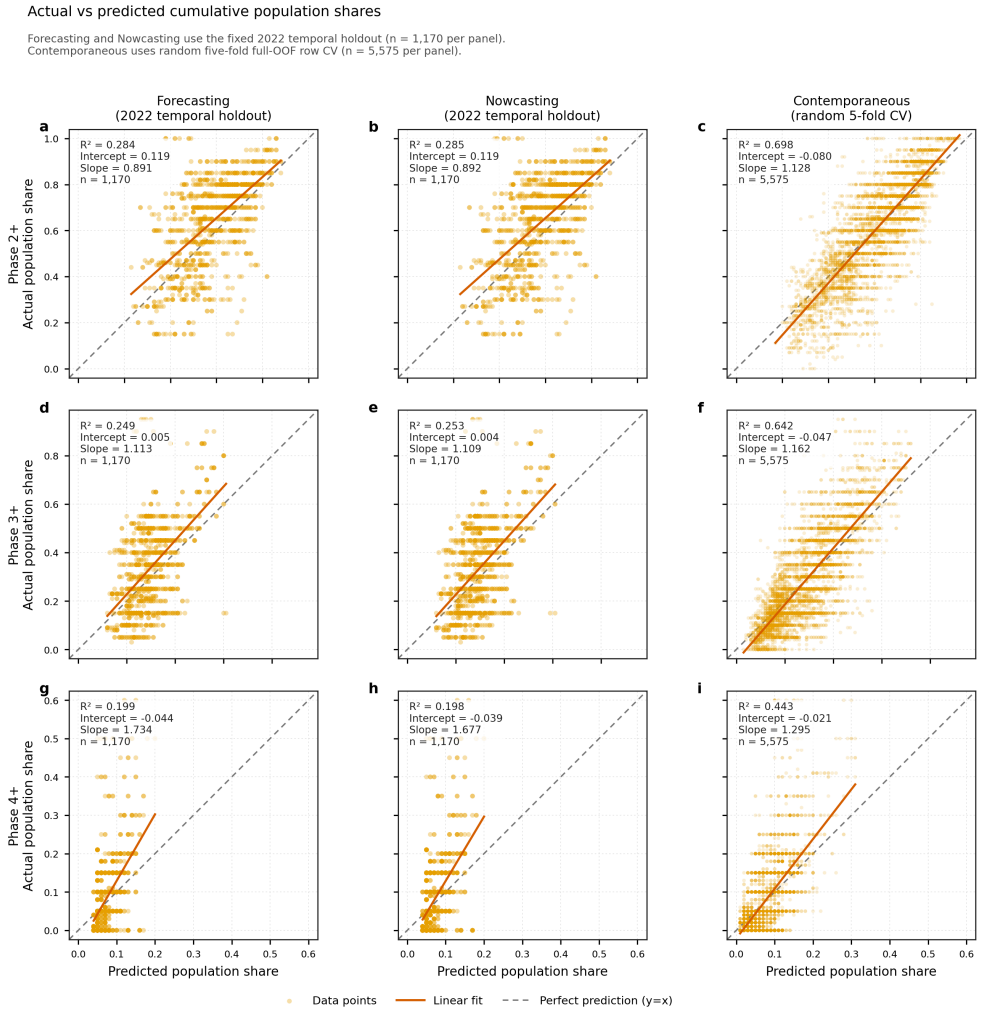


Supplementary Fig. 3 Temporal Validation for Forecasting Hyperparameter Selection
Notes: Forecasting hyperparameters were selected using four expanding-window temporal validation folds restricted to 2017–2021, adapted from [8]. Training began with 2017, followed by validation in 2018; subsequent folds expanded training annually and validated in 2019, 2020 and 2021. The 2022 observations were excluded from every tuning fold. For final forecasting and nowcasting evaluation, observations from 2017–2021 formed the training set and 2022 observations formed the fixed held-out test set. Nowcasting and contemporaneous models were evaluated with fixed hyperparameters, without separate hyperparameter searches.

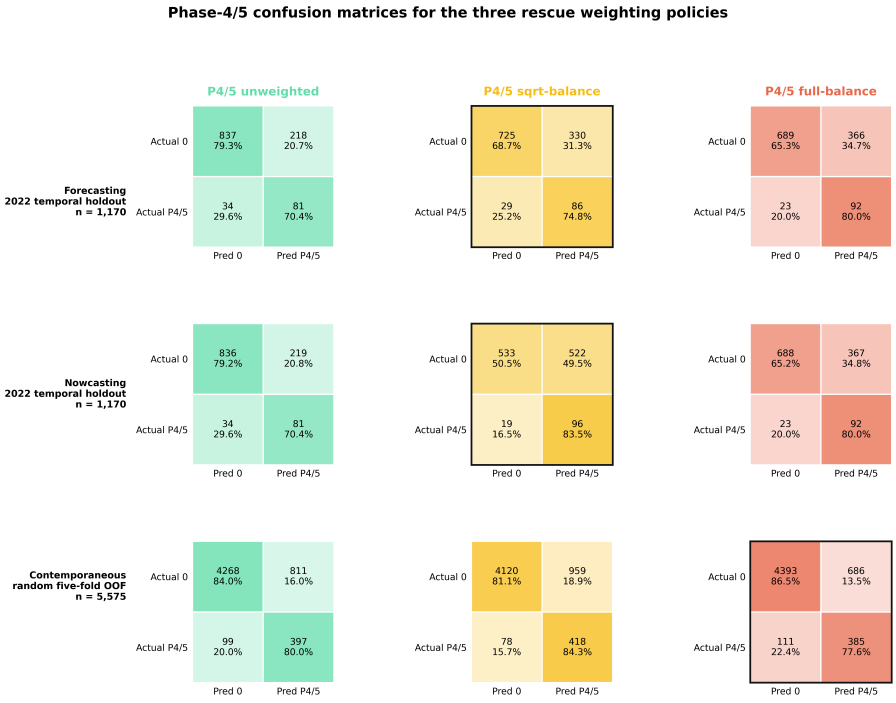


Same observations and five actual phases before/after rescue. Right panels combine predicted P4 and P5;
rescue promotes eligible P3 predictions to P4/5 and does not distinguish P4 from P5.
Cells show count and percentage within each actual phase. Forecasting and Nowcasting use the fixed 2022 temporal holdout (n = 1,170 each).
Contemporaneous uses seed-0 random row-level five-fold full OOF (n = 5,575); its threshold is pooled-OOF selected (not nested/independent).
Protocols and populations differ across rows; cross-row performance is not directly comparable.

Supplementary Fig. 4 IPC Predictions Before and After Combined Phase 4/5 Rescue
Notes: For the contemporaneous panels, actual phases are reconstructed from the observed cumulative population shares using the model's 20% threshold rule, rather than taken from the source-table phase labels used in Fig. 4. Rows show forecasting (a–b), nowcasting (c–d), and contemporaneous modeling (e–f). Left panels retain five predicted IPC classes. Right panels display four predicted categories (P1, P2, P3, combined P4/5), summing the stored P4 and P5 columns. Both sides retain all five actual classes and identical observations. Rescue promotes eligible baseline P3 predictions to combined P4/5; it does not distinguish P4 from P5 or establish improved full five-class classification. Cells report counts and percentages within each actual phase; shading represents the same within-class share. Forecasting and nowcasting use the square-root-balance policy on the 2022 temporal holdout ($n = 1{,}170$ each). Contemporaneous uses the full-balance policy and seed-0 random five-fold row-level OOF predictions ($n = 5{,}575$), with its threshold selected from pooled OOF results. Protocols and populations differ across rows, so cross-row performance is not directly comparable.



Supplementary Fig. 5 Actual and Predicted Cumulative IPC Population Shares
Notes: These diagnostics assess cumulative population-share predictions, not probabilities that an area experiences a food crisis. Columns show forecasting, nowcasting, and contemporaneous estimates. Rows show IPC Phase 2+ (panels a–c), Phase 3+ (panels d–f), and Phase 4+ (panels g–i). Each point represents one observation, with predicted share on the horizontal axis and observed share on the vertical axis. Orange lines regress observed shares on predicted shares; dashed gray lines show equality ($y = x$). Intercept 0 and slope 1 are the linear-calibration reference. Reported R^2 compares the original share predictions with observations, rather than measuring the orange line's fit. Forecasting and nowcasting use the 2022 temporal holdout ($n = 1{,}170$ per panel); contemporaneous estimates use random five-fold OOF predictions ($n = 5{,}575$ per panel). Statistics are descriptive and not directly comparable across columns because evaluation protocols and populations differ.



Supplementary Fig. 6 Binary Confusion Matrices for the Phase 4/5 Rescue Models
Notes: The matrices summarize binary classification performance for identifying observations in IPC Phase 4 or 5 under three class-weighting policies: unweighted, square-root balance, and full balance. Rows correspond to forecasting, nowcasting, and contemporaneous models, while columns correspond to the three weighting policies. "Actual 0" denotes observations below Phase 4, and "Actual P4/5" denotes observations in Phase 4 or 5. Each cell reports the observation count and the percentage within the corresponding actual-class row. Black borders identify the primary policy selected separately for each modeling task using out-of-fold (OOF) results. Forecasting and nowcasting are evaluated on the fixed 2022 temporal holdout ($n = 1{,}170$ each), whereas the contemporaneous model uses seeded random five-fold, row-level OOF predictions ($n = 5{,}575$), with its threshold selected from the pooled OOF predictions. Because the validation protocols and evaluation populations differ, performance should not be compared directly across modeling tasks.

Supplementary Tables

Supplementary Table 1 Model Performance Relative to Benchmark Models

Model	Accuracy	Sensitivity	Precision	R^2
Panel A: Persistence baseline				
Forecasting	0.510	0.612	0.853	–
Nowcasting	0.588	0.745	0.745	–
Panel B: Ordered Probit				
Forecasting	0.491	0.613	0.793	–
Nowcasting	0.490	0.564	0.825	–
Contemporaneous	0.654	0.792	0.840	–
Panel C: OLS				
Forecasting	0.533	0.689	0.784	-0.796
Nowcasting	0.545	0.694	0.794	-0.930
Contemporaneous	0.639	0.742	0.888	0.443
Panel D: ML comparable to Panel A: persistence baseline				
Forecasting	0.672	0.918	0.804	0.243
Nowcasting	0.675	0.914	0.811	0.254
Panel E: ML comparable to Panels B and C				
Forecasting	0.650	0.941	0.775	0.249
Nowcasting	0.654	0.942	0.777	0.253
Contemporaneous	0.693	0.904	0.797	0.642

Notes: This table compares the machine-learning models with persistence, ordered probit, and OLS benchmarks. Accuracy evaluates overall IPC phase classification; sensitivity and precision evaluate food-crisis classification; while out-of-sample R^2 evaluates prediction of the population share in food crisis (Phase 3 or above).

Supplementary Table 2 Model Performance for Alternative Spatial Strategies

Model	Accuracy	Sensitivity	Precision	R^2
Panel A: Exclusion of latitude and longitude				
Forecasting	0.674	0.954	0.789	0.217
Nowcasting	0.676	0.954	0.794	0.223
Contemporaneous	0.686	0.906	0.789	0.628
Panel B: 5-nearest-neighbor spatial averages				
Forecasting	0.671	0.945	0.796	0.134
Nowcasting	0.674	0.944	0.803	0.141
Contemporaneous	0.700	0.912	0.804	0.663
Panel C: 200-km spatial averages				
Forecasting	0.666	0.935	0.802	0.131
Nowcasting	0.673	0.936	0.804	0.136
Contemporaneous	0.698	0.906	0.804	0.680

Notes: This table reports robustness checks using alternative spatial specifications: excluding latitude and longitude, replacing coordinates with five-nearest-neighbor spatial averages of predictors, and using spatial averages within 200 km. Accuracy evaluates overall IPC phase classification; sensitivity and precision evaluate food-crisis classification; while out-of-sample R^2 evaluates prediction of the population share in food crisis (Phase 3 or above).

Supplementary Table 3 Generalization to New Geographic Areas

Model	Accuracy	Sensitivity	Precision	R^2
Panel A: Leave-One-Country-Out				
Forecasting	0.652	0.949	0.774	-0.014
Nowcasting	0.653	0.950	0.777	-0.007
Contemporaneous	0.566	0.918	0.666	0.118
Panel B: Five-Fold Area-Level Cross-Validation				
Forecasting	0.649	0.908	0.798	0.165
Nowcasting	0.656	0.914	0.802	0.170
Contemporaneous	0.830	0.923	0.745	0.718

Notes: This table reports model performance under two validation exercises designed to assess generalization to previously unseen geographic areas. Panel A reports leave-one-country-out results and Panel B reports five-fold area-level cross-validation results. Accuracy evaluates overall IPC phase classification; sensitivity and precision evaluate food-crisis classification; while out-of-sample R2 evaluates prediction of the population share in food crisis (Phase 3 or above).

Supplementary Table 4: Base Variables and Data Sources by Variable Group

Variable Name	Dataset Source
Panel A: Physical Geography	
Elevation	ERA5-Land
Ruggedness	Nunn, Nathan, and Diego Puga (2012)
Slope	ESA
Distance to rivers	World Bank and Andreadis et al. (2013)
Panel B: Conflict	
Event count of battles	ACLED
Event count of explosions	ACLED
Event count of violence against civilians	ACLED
Fatalities of battles	ACLED
Fatalities of explosions	ACLED
Fatalities of violence against civilians	ACLED
Reverse distance weight of fatalities in neighboring 5 stations	ACLED
Reverse distance weight of fatalities in neighboring 10 stations	ACLED
Reverse distance weight of event counts in neighboring 5 stations	ACLED
Reverse distance weight of event counts in neighboring 10 stations	ACLED
Panel C: Weather Conditions	
Rainfall	CHIRPS
2m temperature	ERA5-Land
Standard deviation of 2m temperature	ERA5-Land
Surface solar radiation downwards	ERA5-Land
Standard deviation of surface solar radiation downwards	ERA5-Land
Panel D: Food Prices	
Distance to nearest food market	FAO FPMA
Food price index (reference-price normalized)	FAO FPMA
WFP price change percentage (monthly)	WFP
WFP spatial food price dispersion(monthly)	WFP

Continued on next page

Supplementary Table 4 (continued)

Variable Name	Dataset Source
Panel E: Economic	
Estimated population	IPC
CPI (2010 = 100)	WBG (FP.CPI.TOTL)
GDP per capita	WBG
Gini coefficient	WBG
Control of corruption: percentile rank	WBG
LSCI infrastructure score	WBG (IS.SHP.GCNW.XQ)
VIIRS nightlight mean	VIIRS
Standard deviation of VIIRS nightlight	VIIRS
Market access	Market Access (Weiss, 2015)
Panel F: Agriculture and Land Productivity	
Agricultural ecological zone	Tricht et al. (2023)
GOSIF GPP	GOSIF-GPP
Standard deviation of GOSIF GPP	GOSIF-GPP
Soil volumetric water	ERA5-Land
Standard deviation of soil volumetric water	ERA5-Land
Enhanced vegetation index	NASA
Cropland mask	ASAP (European Commission JRC)
Rangeland mask	ASAP (European Commission JRC)
Volumetric fraction of coarse fragments	ISRIC
Total nitrogen	ISRIC
Soil pH value	ISRIC
Soil organic carbon content in the fine earth fraction	ISRIC
Cation exchange capacity of the soil	ISRIC

Supplementary Notes

We report the hyperparameter search space, temporal validation procedure, model-selection criterion, and final effective parameter values here. We also describe the treatment of missing predictor values in the ML model and in the two benchmark models, ordered probit and OLS.

Hyperparameter tuning

Hyperparameter tuning was performed for forecasting. Nowcasting and contemporaneous models were subsequently evaluated with fixed hyperparameters, without separate hyperparameter searches. The forecasting search exhaustively enumerated the Cartesian product of the eight hyperparameter grids shown below; it was not based on random sampling. Tuning was conducted in two stages. We first selected the parameters for the Phase 3+ cumulative target. In the second stage, the Phase 3+ component was held fixed while parameter settings were selected for the Phase 2+, Phase 4+, and Phase 5 cumulative targets. We prioritized Phase 3+ because it defines food-crisis conditions requiring urgent response under the IPC framework, making both the population share in Phase 3+ and the identification of areas classified as Phase 3+ particularly policy relevant. All 11,664 candidate configurations were evaluated in each stage, corresponding to 23,328 candidate-configuration evaluations across the two searches. We searched all the combinations as it was computationally feasible in this situation and more efficient search strategies could be employed for larger search spaces.

Temporal validation and selection criterion

Each candidate configuration was evaluated using four expanding-window temporal validation folds: 2017 was used to train the first fold and 2018 for validation; 2017–2018 were used for training and 2019 for validation; 2017–2019 were used for training and 2020 for validation; and 2017–2020 were used for training and 2021 for validation. The 2022 data were reserved exclusively for final holdout evaluation. Candidate configurations were ranked by their mean validation score across the four folds:

$$\text{Selection score} = \text{Accuracy} + \text{Precision} + 5 \times \text{Sensitivity} + R^2_{\text{Phase 3+}}$$

The score combines accuracy, precision, sensitivity, and the R2 for the Phase 3+ population share. We assign greater weight to sensitivity because failing to identify true food-crisis cases is of particular concern in an early-warning setting.

Final Effective Parameter Values

Supplementary Table 5 reports the hyperparameter search space. The final effective parameter values used to produce the reported predictions are listed below. The two layers of the nowcasting model used the same parameter settings. The contemporaneous model used a common parameter set for all four cumulative targets. For the final forecasting and nowcasting models, the Phase 2+ cumulative target used `max_depth` =

11, n_estimators = 200, learning_rate = 0.10, subsample = 1.0, colsample_bytree = 0.5, gamma = 0, min_child_weight = 0, and scale_pos_weight = 1. The Phase 3+, Phase 4+, and Phase 5 cumulative targets used the same parameter set: max_depth = 9, n_estimators = 200, learning_rate = 0.10, subsample = 0.5, colsample_bytree = 0.7, gamma = 0.1, min_child_weight = 0, and scale_pos_weight = 1. For the contemporaneous benchmark, all four cumulative targets (Phase 2+, Phase 3+, Phase 4+, and Phase 5) used max_depth = 3, n_estimators = 200, learning_rate = 0.10, subsample = 1.0, colsample_bytree = 0.7, gamma = 0.1, min_child_weight = 0, and scale_pos_weight = 1. All other parameters were left at the corresponding XGBoost defaults. The released reproducible implementation fixes the estimator random state at 0.

Missing values

Our models include 172 predictors, many of which contain missing values. Patterns of missingness may vary systematically across countries, data sources, and conflict contexts.[5] XGBoost handles missing predictor values natively by learning a default direction for missing observations at each tree split. In contrast, OLS and ordered probit require finite design matrices, so their performance may depend on the chosen missing-data treatment.

For the ordered probit and the OLS models reported in Supplementary Table 1, we use feature-wise median imputation estimated from the training data only, followed by standardization using the corresponding training-sample moments. These preprocessing parameters are then applied unchanged to the test data. No observations are dropped, so all models are evaluated on the same 4,405/1,170 temporal training/test cohorts and prespecified splits. The contemporaneous analysis likewise retains all 5,575 observations under fixed five-fold assignments.

We conduct two additional missing-data robustness checks. First, we re-estimate the OLS and ordered probit benchmarks after adding a binary missingness indicator for each predictor; these results are reported in Panels A and B in Supplementary Table 6. Second, we repeat the model comparisons after excluding the 10% most-missing predictors and, separately, the three countries with the highest missingness rates. These results are reported in Panels C–F, with the corresponding ML results shown in Panels G and H of Supplementary Table 6.

[5] Using all 5,575 observations, we find that conflict-affected observations have, on average, 2.92 fewer missing forecasting predictors and 2.11 fewer missing nowcasting predictors than non-conflict observations. Inference clustered at the area level indicates that these differences are statistically significant, although modest in magnitude.

Supplementary Table 5
Hyperparameter Search Space

Hyperparameters	Search Space
max_depth	3, 7, 9, 11
n_estimators	200, 500, 800
learning_rate	0.01, 0.05, 0.08, 0.10
subsample	0.5, 0.7, 1.0
colsample_bytree	0.5, 0.7, 1.0
gamma	0, 0.1, 0.2
min_child_weight	0, 1, 2
scale_pos_weight	1, 2, 3

Supplementary Table 6: Robustness Check: Model Performance Relative to Benchmark Models

Model	Accuracy	Sensitivity	Precision	R^2
Panel A: Ordered Probit (+ missingness indicators)				
Forecasting	0.519	0.642	0.798	–
Nowcasting	0.551	0.753	0.795	–
Contemporaneous	0.663	0.805	0.845	–
Panel B: OLS (+ missingness indicators)				
Forecasting	0.544	0.712	0.792	-0.571
Nowcasting	0.603	0.877	0.760	-0.496
Contemporaneous	0.646	0.887	0.752	0.466
Panel C: Ordered Profit (drop 10% most-missing predictors)				
Forecasting	0.498	0.610	0.801	
Nowcasting	0.507	0.613	0.804	
Contemporaneous	0.647	0.785	0.836	
Panel D: OLS (drop 10% most-missing predictors)				
Forecasting	0.534	0.689	0.788	-0.743
Nowcasting	0.546	0.699	0.801	-0.756
Contemporaneous	0.642	0.888	0.748	0.439
Panel E: Ordered Probit (drop 3 most-missing countries)				
Forecasting	0.489	0.608	0.794	
Nowcasting	0.488	0.544	0.843	
Contemporaneous	0.655	0.843	0.792	
Panel F: OLS (drop 3 most-missing countries)				
Forecasting	0.531	0.686	0.783	-0.792
Nowcasting	0.543	0.689	0.796	-0.931
Contemporaneous	0.642	0.890	0.744	0.443
Panel G: ML comparable to Panel A, B, C & D				
Forecasting	0.650	0.941	0.775	0.249
Nowcasting	0.654	0.942	0.777	0.253
Contemporaneous	0.693	0.904	0.797	0.642
Panel H: ML comparable to Panel E & F				
Forecasting	0.676	0.950	0.794	0.294
Nowcasting	0.679	0.942	0.803	0.303
Contemporaneous	0.690	0.905	0.794	0.637

Notes: This table compares the machine-learning models with persistence, ordered probit, and OLS benchmarks. Accuracy evaluates overall IPC phase classification; sensitivity and precision evaluate food-crisis classification; while out-of-sample R^2 evaluates prediction of the population share in food crisis (Phase 3 or above).